

Midterm Sample

ECON 441: Introduction to Mathematical Economics

Instructor: Div Bhagia

Print Name: _____

This is a closed-book test. You may not use a phone or a computer.

Time allotted: 70 minutes

Total points: 20

Please show sufficient work so that the instructor can follow your work.

I understand and will uphold the ideals of academic honesty as stated in the honor code.

Signature: _____

1. (5 pts) Answer the following questions.

(a) (1 pt) Consider a mapping $f(x)$. For two distinct values of x , x_0 and x_1 , $f(x_0) = f(x_1)$. Is f a valid function? Answer yes or no.

(b) (1 pt) Consider two sets A and B , where A is the set of all odd real numbers and B is the set of all even real numbers. What is the union of A and B ?

(c) (1 pt) Find the union and intersection for the following sets:

$$A = \{x : x \text{ is an even number}\} \quad B = \{2, 4, 8\}$$

(d) (1 pt) Does the inverse of the function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$ exist? If yes, find the inverse.

(e) (1 pt) Consider the following two-variable function:

$$f(x, y) = x + y$$

where $x \in (0, 1)$ and $y \in (0, 1)$. What is the range of f ?

2. (5 pts) Consider the following matrix

$$A = I - X(X^T X)^{-1} X^T$$

(a) (3 pts) Is A a square matrix? Show your work or reasoning that led you to this conclusion.

(b) (2 pts) Prove that A is idempotent i.e. $AA = A$.

3. (10 pts) An economy has three sectors: agriculture, manufacturing, and services. Denote their outputs (in billions of dollars) by x , y , and z , respectively. The planning targets require that agricultural output exceed twice the services output by 2, that manufacturing and services output together equal 12, and that total output across the three sectors equal 24:

$$x - 2z = 2$$

$$y + z = 12$$

$$x + y + z = 24$$

- (a) (1 pt) Write this system of equations in matrix format i.e.,

$$Av = b$$

What is A , v , and b equal to?

(b) (3 pts) Calculate the adjoint of A .

(c) (2 pts) Calculate the determinant of A . Is A nonsingular?

(d) (1 pt) If you premultiply A^{-1} on both sides of the equation $Av = b$, you should be able to derive an expression to solve for v . Write down this expression.

(e) (3 pts) Using the expression in (d), solve for v^* , i.e., the output of each sector.